Machine Learning for Finance

Instructor: Chao ZHANG
TA: Zhanghao Chen

Jan 27, 2026

Course Description

Machine Learning

Nuanced Differences in Academic Finance Fields

ML4Finance in Industry

Course Agenda

Course Overview

This course explores the intersection of machine learning (ML) and finance, with applications spanning asset pricing, portfolio optimization, risk management, and more.

Students will learn how to apply state-of-the-art machine learning techniques to financial data and gain practical experience through hands-on projects.

Prerequisites:

- Background in finance or economics
- Familiar with machine learning, applied statistics, and programming (Python)

Class Information: Jan 23 to May 11, every Tuesday, 9:00-11:50 am, Rm202, E4.

Office Hour: Email Appointment, chaoz@hkust-gz.edu.cn

Suggested Readings

1. *Financial Machine Learning*, by Kelly et al. [2023]
2. *Advances in Financial Machine Learning*, by De Prado [2018]
3. *Recent Advances in Reinforcement Learning in Finance*, by Hambly et al. [2023]
4. *The Elements of Financial Econometrics*, by Fan and Yao [2017]
5. *The Elements of Statistical Learning*, by Hastie et al. [2017]

Homework

- **Assignment 1: Linear Regression and Stock Returns**
  - **Task**: Apply regression to predict stock returns using U.S. stock market data (provided). Test understanding of linear regression, its strengths, and limitations.
- **Assignment 2: High-Frequency Data Modeling**
  - **Task**: Replicate and Improve (optional) the DeepLOB model from Zhang et al. [2019]. Modeling high-frequency data (provided) in financial markets.

Submissions:

- All assignments must be done individually and submitted in Jupyter Notebook.
- Deadline: Two weeks.
- Late submission: 75% on day 1; 50% on day 2; 25% on day 3.
- No submissions accepted after 3 days without an extension.

Final Project

- **Option 1: Replicate Jiang et al. [2023]**
  - **Data**: Historical financial data provided.
  - **Outcome**: Predict stock returns in multiple horizons based on OHLC chart data.
- **Option 2: Statistical Arbitrage via LOB Data**
  - **Data**: Raw LOBSTER data provided.
  - **Outcome**: Develop a new statistical arbitrage strategy using machine learning.
- **Propose Your Own Project: Leveraging LLMs in Finance**
  - Automated financial news summarization.
  - Generate financial reports.
  - Sentiment analysis for stock prediction.
  - Other LLM-based applications in finance.

Final Project

Submission Requirements:

- Deadline: May 18
- A final report using the **NeurIPS template**.
- Teams of up to ≤ 3 members.
- Clear distribution of responsibilities among team members.
- Include all code used in the project.

Evaluation Criteria:

- Problem Definition (10%): Clearly defined and well-structured research question or problem.
- Methodology (30%): Justification and rationale behind the model choices, including any assumptions.
- Implementation (30%): Quality, efficiency, and readability of the code, with appropriate comments and documentation.
- Analysis & Results (20%): Depth of the analysis, quality of results, and insightful interpretation of the findings.
- Presentation (10%): Clarity, organization, and professionalism of the final presentation.

Grading

Course Grading Breakdown

1. Participation: 5%
2. Course Assignment I: 20%
3. Course Assignment II: 25%
4. Final Project + Presentation: 50%

Project I: NLP-Measured Political Risk and Volatility

Chris Li, Derrick Wang, Shitong Xiao, Colbert Yu

This paper builds a new embedding-based firm-level political risk measure using S&P 500 earnings calls. It embeds policy-relevant sentences and constructs the risk score from each sentence's cosine similarity to predefined uncertainty prototypes net of its similarity to certainty/irrelevance prototypes. The measure lines up well with major political episodes and generates plausible sector exposure patterns. The team further shows that their measure exhibits a stronger contemporaneous association with both realized and implied volatility than the dictionary-based measure. Finally, they find that adding their political risk measure to an existing volatility forecasting model reduces out-of-sample forecast loss, while the inclusion of the dictionary-based measure worsens forecast accuracy.

Paper in one picture/table:

Table 5: HMSE of Volatility Forecasting Models

This table reports the HMSE from estimating HAR-RV model under different specifications. The dependent variable is the simple average of realized volatility from day $t+1$ to $t+h$. The predictors in Column (1) include realized volatility at day t ($RV_{i,t}$), lagged weekly realized volatility ($RV_{i,w,t}$), and lagged monthly realized volatility ($RV_{i,m,t}$). The predictors in Column (2) include $RV_{i,t}$, $RV_{i,w,t}$, $RV_{i,m,t}$, and the value of the political risk measure from Hassan et al. (2019) constructed from the most recent quarterly earnings conference call held on or before day t for firm i ($PRisk_{\text{ref}}$). The predictors in Column (3) include $RV_{i,t}$, $RV_{i,w,t}$, $RV_{i,m,t}$, and the value of the our newly constructed political risk measure from the most recent quarterly earnings conference call held on or before day t for firm i ($PRisk_{\text{emb}}$). N-days reports the number of unique trading dates in the out-of-sample period. All PRisk measures are winsorized at the first and last percentile.

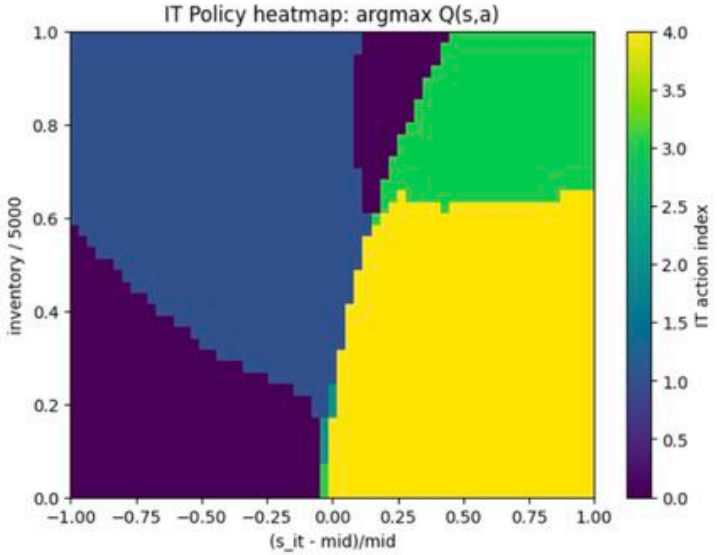
	w/o PRisk	**with PRisk$_{ref}$**	**with PRisk$_{emb}$**
	(1)	(2)	(3)
$h=1$	2,653.20	3,318.85	2,593.21

Project II: Reinforcement Learning in Simulated Financial Markets

Peng Liu, Xiangyu Sun, Yiqiao Sun, Yize Wang

This paper builds a simulated limit-order-book market in which reinforcement learning agents trade a short-lived asset. The model helps reveal how private information, inventory risk, and competition shape price discovery and profits for AI trading agents. In the single-market-maker environment, the team shows that the RL-informed trader endogenously learns to hide its orders when prices are far from its private signal, to condition on recent price trends, and to exploit a mechanically adaptive but non-learning market maker. When two RL market makers compete, they examine how their learned quoting rules affect price efficiency and the detectability of informed trading from public data, connecting the AI-collusion literature to the concrete structure of LOB markets.

Paper in one picture/table:



Project IV: Trading Volume Alpha with Volatility-Aware Execution

Alex Carr, Stan Nicole, Christelle Saad, Drumil Trivedi

This paper extends the trading volume alpha framework of Goyenko et al. (2025) to make trade execution decisions depend on both volume and volatility. Using linear (Ridge) and neural network (RNN-LSTM) models to forecast daily volume surprises and plugging them into an economic loss function, the team finds that economically fine-tuned RNNs work best for large, liquid stocks, while simple linear models are superior for small caps. Allowing price impact to scale with volatility surprises yields modest gains for large caps but substantially cuts economic loss—by about 30–40%—for illiquid stocks in high-volatility regimes relative to volume-only RNNs.

Paper in one picture/table:

Table 7: Smallest Universe — MEL Reduction in High Volatility Regime

Method	**$100M**	**$1B**	**$10B**	**$100B**
MA5	0.0%	0.0%	0.0%	0.0%
Linear Models				
olstech	+5%	+6%	+6%	+6%
ols.vol_adj tech	+27%	+29%	+32%	+24%
Feedforward Neural Networks				
nntech	+8%	+9%	+10%	+7%
nn.vol_adj tech	+30%	+32%	+36%	+21%
nnall	+7%	+8%	+9%	+6%
nn.vol_adj all	+28%	+30%	+34%	+19%
Recurrent Neural Networks				
rnntech	+10%	+11%	+12%	+8%
rnn.vol_adj tech	+32%	+34%	+39%	+22%
rnnall	−12.3%	−12.3%	−7.9%	−11.3%
rnn.vol_adj all	+29%	+30%	+32%	−13%

Course Description

Machine Learning

Nuanced Differences in Academic Finance Fields

ML4Finance in Industry

Course Agenda

What is Machine Learning?

- Machine Learning (ML) $\Leftrightarrow$ Artificial Intelligence.
- ML models improve performance by identifying patterns in data.
- ML algorithms use data to make predictions or decisions, refining their predictions as more data is provided.
- **Supervised learning**: The model is trained on labeled data (input-output pairs) and learns to predict the output for new, unseen inputs.
- **Unsupervised learning**: The model is given unlabeled data and should find hidden structures within the data.
- **Reinforcement learning**: The model learns by interacting with an environment and receiving feedback in the form of rewards, improving its behavior over time.

What is Machine "Learning"?

1. **Data Exposure:** The model is provided with data.
   - images, science, finance, economics, etc.
2. **Specific Task:** Learning the right problem!
   - Formalizing the task and aligning the solution with intuition.
   - Risk-averse investor: mean-variance utility
   - Highly correlated assets should be avoided.
3. **Learning Algorithm:** determines how the model will learn from data.
   - Gradient descent
   - Transfer learning: a model trained on one task (like Pretrained LLMs) can be fine-tuned for other tasks (e.g., text classification).
   - Few-shot learning: LLMs are capable of performing tasks with minimal examples.
4. **Improvement over Time:** As the model receives more data, it continually refines its predictions.

Course Description

Machine Learning

Nuanced Differences in Academic Finance Fields

ML4Finance in Industry

Course Agenda

Field	Primary Emphasis	Typical Role of ML
Math Finance	Theory-driven modeling of prices and risks	Approximation, control, numerical solver, deep hedging
Quant Finance	Data-driven trading, forecasting, and risk	Prediction and decision-making
Fin. Engineering	Large-scale systems and algorithmic design	ML as system optimizer and representation learner
Fin. Economics	Economic mechanisms, causal inference, interpretability	Measurement, prediction, and heterogeneity

- Mathematical Finance: **Is it well-defined?**
- Quantitative Finance: **Does it work?**
- Financial Engineering: **How should the system be built?**
- Financial Economics: **Why does it happen?**

Course Description

Machine Learning

Nuanced Differences in Academic Finance Fields

ML4Finance in Industry

Course Agenda

ML4Finance in Industry

- Typical Application Scenarios
  - **Quant Trading:** HFT, algorithmic execution, market making
  - **Investment:** Factor discovery, alpha mining, portfolio
  - **Banking:** Credit scoring, fraud detection, anti-money laundering
  - **Analytics:** NLP on earnings calls, sentiment analysis, alternative data
- **Industries Covered**
  - Asset management, hedge funds, proprietary trading firms
  - Investment banks and broker-dealers
  - Commercial banks, consumer finance, and Fintech
  - Insurance, exchanges, and regulatory bodies (RegTech)
- **Roles Across the Organization**
  - Front Office: Alpha research, pricing, execution, robo-advisory
  - Middle Office: Market/Credit risk, liquidity monitoring, capital allocation
  - Back Office: Surveillance, compliance, operations automation, settlements

Course Description

Machine Learning

Nuanced Differences in Academic Finance Fields

ML4Finance in Industry

Course Agenda

Lecture 1 (Jan 27): Introduction to Financial Machine Learning

- Overview
- Prices are Predictions
- Information Set
- Functional Forms
- Challenges of Applying Machine Learning in Finance

Lecture 2 (Feb 3): Empirical Asset Pricing 1

- Introduction to Financial Data
- Experimental Design
- Simple Linear Models
- Penalized Linear Models
- Dimension Reduction
- HW1

Lecture 3 (Feb 10): Empirical Asset Pricing 2

- Decision Trees
  - Ensembling method: bias, var, inherent noise
  - Bryzgalova et al. [2019], Cong et al. [2022]
- Neural Networks (Gu et al. [2020], Choi et al. [2025])
- Alternative Data
  - Image (Jiang et al. [2023])
  - Text (Bybee et al. [2024])
  - Sound

Lecture 4 (Feb 24): High-Frequency Return Prediction

- Financial Time Series Forecasting
- Limit Order Book
- Sampling Frequency (Ait-Sahalia et al. [2025])
- Order Flow (Kolm et al. [2023])
- Universality (Sirignano and Cont [2019])
- Cross Impact (Chinco et al. [2019], Cont et al. [2023])
- HW2

Lecture 5 (March 3): Optimal Portfolios

- "Plug-in" Portfolios
- Integrated Estimation and Optimization
- Maximum Sharpe Ratio Regression
- Accounting for Trading Costs (Goyenko et al. [2024])

Lecture 6 (March 10): Reinforcement Learning in Finance

- Introduction to Reinforcement Learning
- Applications in Finance
  - optimal execution
  - market making
- Selected papers about RL Applications (Hambly et al. [2023])
- Invited Speaker

Lecture 7 (March 17): Generative AI for Synthetic Financial Data

- Introduction to Generative AI
- Applications in Finance
- Backtesting
- Challenges of Generative AI in Finance
- QuantGAN (Wiese et al. [2020], TailGAN (Cont et al. [2025])

Lecture 8 (March 24): Risk Management with Graphs

- Graph-based ML
- Market Risk:
  - Value-at-Risk (VaR)
- Credit Risk:
  - Fraud Detection
  - Anomaly detection techniques
  - Predicting loan defaults
  - Credit Scoring
- Invited Speaker

Lecture 9 (March 31): Large Language Models (LLM)

- Sentiment Analysis in Finance
- Automated Financial Report Generation
- Customer Service and Chatbots in Financial Institutions
- Reasoning
- Challenges
- Invited Speaker
- Final Project Proposal

Lecture 10 (April 7): Model Interpretability and Ethical Considerations

- Visualization:
  - Cluster Algorithms
  - t-SNE for Financial Data
  - Neo4j Graph Database
- Model Interpretability Methods:
  - LIME and SHAP Explanations

Lecture 11 (April 14): Advanced Topics in Machine Learning for Finance

- Transfer Learning
  - Domain Adaptation
- Federated Learning
  - Privacy-Preserving

Lecture 12 (April 21): Case Studies and Hands-On Projects

- Case Studies
- Hands-on Projects
  - Empirical Asset Pricing via Machine Learning
  - High-Frequency Return Prediction
  - LLM in Finance

Lecture 13 (April 28): Final Project and Presentations

Yacine Ait-Sahalia, Jianqing Fan, Lirong Xue, and Xiaonan Zhu. How and when are high-frequency stock returns predictable? *Management Science*, 2025.

Svetlana Bryzgalova, Markus Pelger, and Jason Zhu. Forest through the trees: Building cross-sections of stock returns. *Available at SSRN 3493458*, 2019.

Leland Bybee, Bryan T Kelly, Asaf Manela, and Dacheng Xiu. Business news and business cycles. Technical report, 2024.

Alex Chinco, Adam D Clark-Joseph, and Mao Ye. Sparse Signals in the Cross-Section of Returns. *Journal of Finance*, 74(1):449–492, 2019.

Darwin Choi, Wenxi Jiang, and Chao Zhang. Alpha go everywhere: Machine learning and international stock returns. *Review of Asset Pricing Studies*, page raaf005, 2025.

Lin William Cong, Guanhao Feng, Jingyu He, and Xin He. Asset pricing with panel tree under global split criteria. 2022.

Rama Cont, Mihai Cucuringu, and Chao Zhang. Cross Impact of Order Flow Imbalance in Equity Markets. *Quantitative Finance*, 2023.

Rama Cont, Mihai Cucuringu, Renyuan Xu, and Chao Zhang. Tail-gan: Learning to simulate tail risk scenarios. *Management Science*, 2025.

Marcos Lopez De Prado. *Advances in financial machine learning*. John Wiley & Sons, 2018.

Jianqing Fan and Qiwei Yao. *The elements of financial econometrics*. Cambridge University Press, 2017.

Ruslan Goyenko, Bryan T Kelly, Tobias J Moskowitz, Yinan Su, and Chao Zhang. Trading volume alpha. *Available at SSRN 4802345*, 2024.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical Asset Pricing via Machine Learning. *Review of Financial Studies*, 33(5):2223–2273, 2020.

Ben Hambly, Renyuan Xu, and Huining Yang. Recent advances in reinforcement learning in finance. *Mathematical Finance*, 33(3):437–503, 2023.

Trevor Hastie, Robert Tibshirani, and Jerome Friedman. The elements of statistical learning: data mining, inference, and prediction, 2017.

Jingwen Jiang, Bryan Kelly, and Dacheng Xiu. (Re-) Imag (in) ing price trends. *The Journal of Finance*, 78(6):3193–3249, 2023.

Bryan Kelly, Dacheng Xiu, et al. Financial machine learning. *Foundations and Trends (R) in Finance*, 13(3-4):205–363, 2023.

Petter N Kolm, Jeremy Turiel, and Nicholas Westray. Deep Order Flow Imbalance: Extracting Alpha at Multiple Horizons from the Limit Order Book. *Mathematical Finance*, 2023.

Justin Sirignano and Rama Cont. Universal Features of Price Formation in Financial Markets: Perspectives from Deep Learning. *Quantitative Finance*, 19(9): 1449–1459, 2019.

Magnus Wiese, Robert Knobloch, Ralf Korn, and Peter Kretschmer. Quant GANs: Deep Generation of Financial Time Series. *Quantitative Finance*, pages 1–22, 2020.

Zihao Zhang, Stefan Zohren, and Stephen Roberts. Deeplob: Deep Convolutional Neural Networks for Limit Order Books. *IEEE Transactions on Signal Processing*, 67(11):3001–3012, 2019.